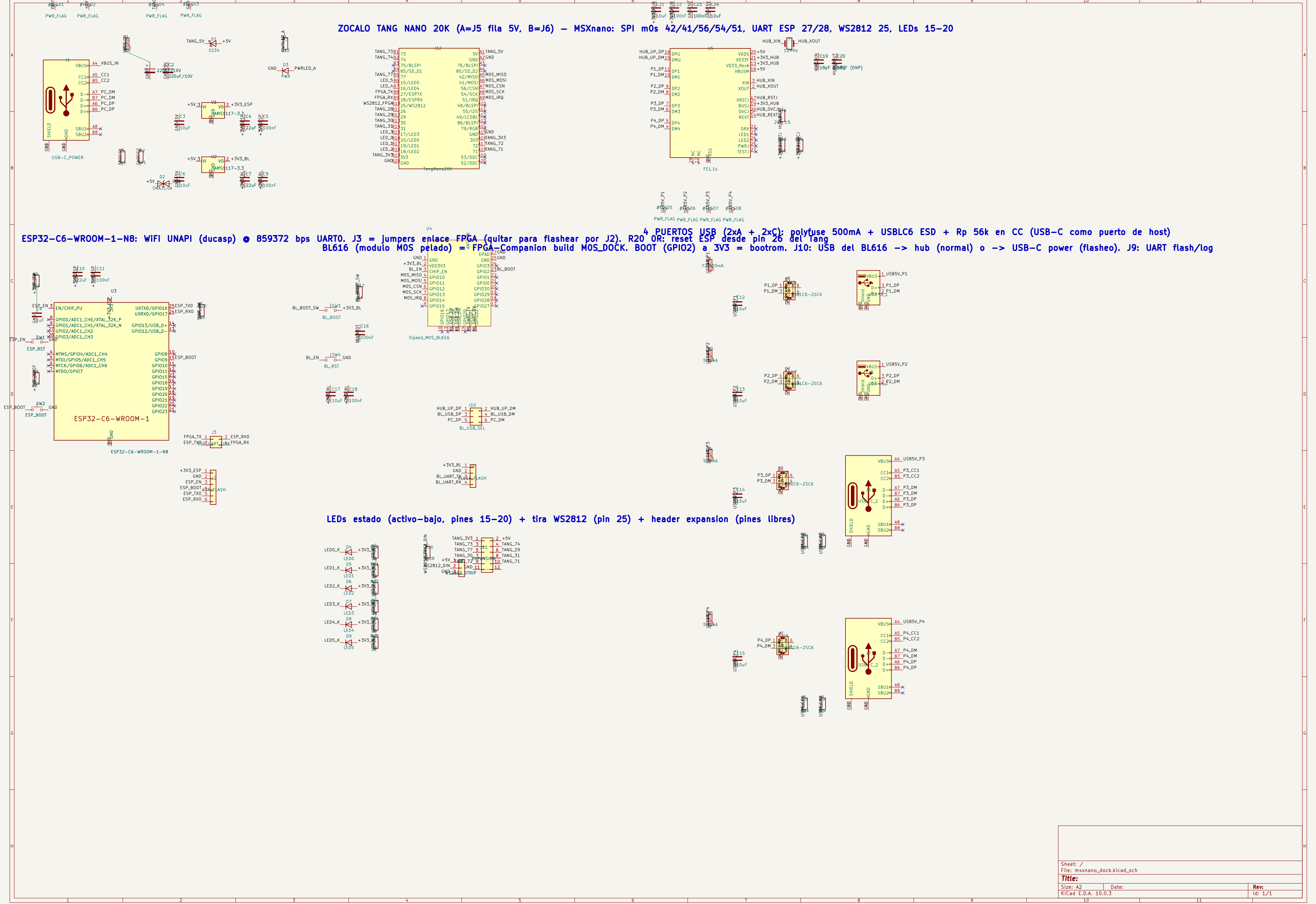


ALIMENTACION: USB-C 5V -> fusible 2A -> +5V | SS34 -> TANG_5V | 2x AMS1117 (ESP / BL616)

HUB USB 2.0 FE1.1s (self-powered, BUSJ=1, OVCJ pull-up) – upstream = BL616 host

ZOCALO TANG NANO 20K (A=J5 fila 5V, B=J6) – MSXnano: SPI m0s 42/41/56/54/51, UART ESP 27/28, WS2812 25, LEDs 15–20



ESP32-C6-WROOM-1-N8: WIFI UNAPI (ducas) ● 859372 bps UART0. J3 = jumpers enlase FPGA (quitar para flashear por J2). R20 OR: reset ESP desde pin 26 del Tang BL616 (modulo MOS pelado) = FPGA-Companion build MOS_DOCK. BOOT (GPIO2) a 3V3 = bootrom. J10: USB del BL616 -> hub (normal) o -> USB-C power (flasheo). J9: UART flash/log

LEDs estado (activo-bajo, pines 15–20) + tira WS2812 (pin 25) + header expansion (pines libres)